Joshua Green

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EDUCATION

PhD – Ocean and Earth Sciences, **October 2022 – Present**
University of Southampton, Southampton, UK
- Research Focus: Compound Flood Modelling, Climate, Hazards, Disasters, Risk, Vulnerability

Bachelor of Science – Geographical Sciences, **Graduated: May 2021**
Co-operative Placement Program
University of British Columbia, Vancouver BC, CAN
- Discipline Focus: Geospatial Sciences, Geomorphology, Environmental Sciences & Policy
- Minor: Geographic Information Science and Geographical Computation

WORK EXPERIENCE

Flood Risk Model Developer **September 2021 – Present**
Fathom Global Ltd, Bristol, UK
- Developed large-scale tropical cyclone and compound flood risk models and vendor CAT risk products to support clients (brokers/underwriters/engineers) in the finance/insurance sector
- Led climate communication and presented vendor risk reports to the public and stakeholders
- Conducted numerical modelling and probabilistic statistical analysis with Python, Matlab, and R
- Curated multi-hazard loss databases and supported exposure and vulnerability modelling

NASA DEVELOP Research Analyst & Team Lead **June – August 2021**
NOAA National Centers for Environmental Information, Asheville NC, US
- Managed a team of 5, coordinating project tasks, and leading communication with local and federal government for agricultural drought monitoring and capacity development in Illinois
- Designed a Python toolset to evaluate the suitability of remotely sensed, modeled, and in-situ soil moisture data products for assessing drought conditions
- Produced soil moisture anomaly and percentile time-series datasets, multidimensional rasters, and statistical products to support decision-makers including the Illinois State Water Survey

NASA DEVELOP Research Analyst **June – August 2020**
NASA Jet Propulsion Laboratory, Pasadena CA, US
- Developed infrastructure and road vulnerability risk reports for permafrost-subsidence in Alaska to support transporation and economic risk management for state and federal government
- Processed and analyzed LiDAR and L & C-band SAR (UAVSAR and Sentinel-1) data, and utilized ArcGis-Pro, QGIS, and Python to assess and map permafrost-induced surface deformation
- Presented findings at NASA Applied Science Week 2020 - *https://youtu.be/4npjdqDdJ4M?t=805*

Transport Canada Environmental Officer **January – June 2020**
Transport Canada, Pacific Environmental Team, Vancouver BC, CAN
- Contributed to Environmental Impact Assessment, contaminated site remediation, and environmental protection projects in British Columbia
- Developed Project-Approval-Document report used for long term project planning
- Authored Environmental Assessment project reports guiding high-level policy meetings
- Drafted annual budgets and processed invoices for contaminated site remediation projects

Environmental Assessment Project Officer **September – December 2019**
BC Environmental Assessment Office, Metal Mining Operations Team, Victoria BC, CAN
- Coordinated and managed operational projects and implementation of new federal EA policy

- Analyzed adverse socio-economic and environmental effects of development projects
- Guided consultations with stakeholders, governments, First Nations, and public on EA processes

Soil Science Research Assistant **May – August 2019**
BC Ministry of Forests, Lands and Natural Resource Operations, Prince George BC, CAN

- Analyzed pre-glacial lake reconstruction in northern BC by integrating LiDAR data and digitization in assessments of the paleo-shoreline landform using Global Mapper
- Generated and formatted stereo LiDAR imagery from point cloud data for 3D vector terrain analysis using DAT/EM Summit Evolution and Global Mapper

NASA Research Intern in the Hydrology Laboratory/USRA **July – September 2018**
NASA Goddard Space Flight Center, Greenbelt MD, US

- Advanced a citizen science-based landslide monitoring project by adding verified content to the Cooperative Open Online Landslide Repository (COOLR) and Global Landslide Catalogue (GLC)
- Designed satellite-based GIS maps to generate landslide susceptibility for trend and change analysis
- Standardized landslide databases using comparative analysis to create a data flow diagram

Forestry Research Assistant (Field and Laboratory) **May – August 2017, May-June 2018**
UBC Faculty of Forestry, Vancouver BC, CAN

- Conducted fieldwork according to standards of the Canadian National Forest Inventory
- Surveyed and photographed field sites and plot boundaries using GPS units and Avenza software
- Evaluated bulk-mass density of soil, wood, and plant samples, and prepared for carbon analysis

CLIMATE/HAZARD PROJECTS

Earth Observations to Enhance Drought Monitoring in Illinois
- *https://jagreen1.github.io/projects/NASA_DEVELOP_IllinoisDisaster2021.html*

Earth Observations to Identify Permafrost Subsidence and Infrastructure Vulnerability in Alaska
- *https://jagreen1.github.io/projects/NASA_DEVELOP_AlaskaTransportation2020.html*

Landslide Susceptibility: Comparison of Global and Regional Models
- *https://jagreen1.github.io/projects/NASA_GSFC_Macedonia_Landslide.html*

Geomorphology in Haida Gwaii: Hillslope-Landslide Area Coupling
- *https://jagreen1.github.io/projects/Geomorphology_Haida_Gwaii.html*

Geographically Weighted Regression Analysis of Wildfires in British Columbia
- *https://jagubc.wixsite.com/bc-wildfire-gwr*

HIGHLIGHTS & QUALIFICATIONS

- Project Management
  - Expertise leading multi-year commercial product development for finance/insurance risk
  - Knowledge of environmental assessment, natural resource, and disaster policy operations
  - Strong service orientation and effective communicator, technical writer, and public speaker
  - Highly successful in coalition building, team collaboration, and commercial business
- Technical
  - Programming with Python, Matlab, R, JavaScript, HTML; and HPC systems
  - Nat CAT risk modelling, exposure mangement, and vulnerability analysis
  - GIS and geospatial analysis using ArcGIS/QGIS/Google Earth Engine/Global Mapper
  - Map and web map design using Adobe Illustrator/Mapbox/Leaflet
  - Relational database applications using SQL/MS Access/Excel
  - Processing and analysis of LiDAR point cloud and SAR data